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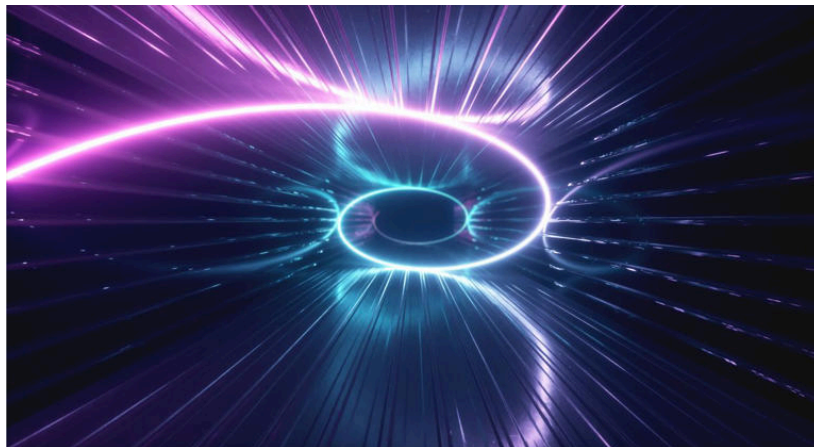
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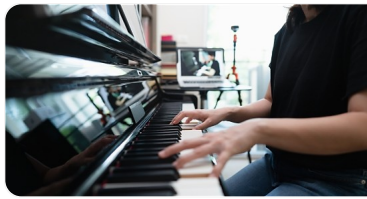


Do things in the quantum world fall in the same way? It appears to be so. - Image Credit: FlashMovie/Shutterstock.com
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An international team of researchers has demonstrated that the equivalence principle, a cornerstone of Einstein's theory of general relativity, holds even in the quantum world. This is a very important result.

Quantum mechanics and general relativity are the foundations of modern physics. The first explains the behavior of atoms and tiny objects, while the second explains how objects fall and gravity shapes the universe.

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Together they allow us to explain so much of the universe. They have been tested time and time again, and yet, combining them to study certain objects and events - like black holes - leaves us with nonsense.

Physicists believe that there's another theory out there that can combine these two into one. We haven't found it yet, but every crumb or hint might take us a step closer.

This latest work sits in that tradition, even if it has confirmed that Einstein's theory is once again right. It all comes down to the weak equivalence principle, also known as the free fall principle.

This states that, in a gravitational field, two objects made of different materials and with different masses fall to the ground with the same acceleration as long as no other forces are acting upon them.

The principle goes all the way back to Galileo, who demonstrated it by dropping cannonballs of different masses from the Leaning Tower of Pisa, which subsequently hit the floor at the same time. More recently, Apollo 15 commander David Scott demonstrated it on the Moon by dropping [a feather](#) and [a hammer](#) and recording them both landing on the surface together.

▶ **Related video:** Quantum discovery at MIT surprises physicists (WooGlobe)



Quantum discovery at MIT surprises physicists

The free fall principle, on a much smaller scale

In the quantum world, the setup was a little bit different. Researchers chilled rubidium atoms to near absolute zero, and then placed them into a superposition state that allowed them to travel along two different paths at once. The atoms can be interpreted as moving in quantum waves.

One part of the atomic wave was then kept stationary by the use of a magnetic field, while the other was first pushed with a different magnetic pulse, and then allowed to fall under gravity.

At the end of the free fall, that part of the atomic wave was then brought back to the position of the part held stationary, at which point the waves interfered with

each other.

This interference allowed the researchers to compare the tiny differences in quantum phase accumulated while one part was in free fall and the other held still. That measurement was consistent with the prediction of Einstein's equivalence principle when applied to a quantum wave, providing an insight into the space where general relativity meets quantum mechanics.



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"We have no consistent theory telling us why quantum physics should fail," Professor Vlatko Vedral, from the department of physics at the University of Oxford, said in a [statement](#). "This experiment pushes quantum mechanics into one of its most intriguing frontiers, gravity, and shows that, once again, its predictions hold."

So, where do quantum mechanics and relativity start to disagree? Another of the study's coauthors, Professor Sir Roger Penrose, has argued that larger objects held in very long superpositions might show the cracks in our theories.

Experiments are now underway to recreate this experiment with nanodiamonds.

A paper describing the results was published in [Science Advances](#).

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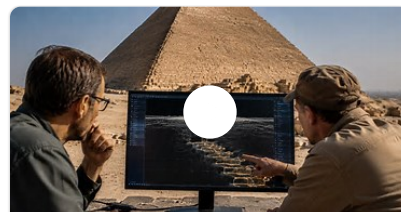
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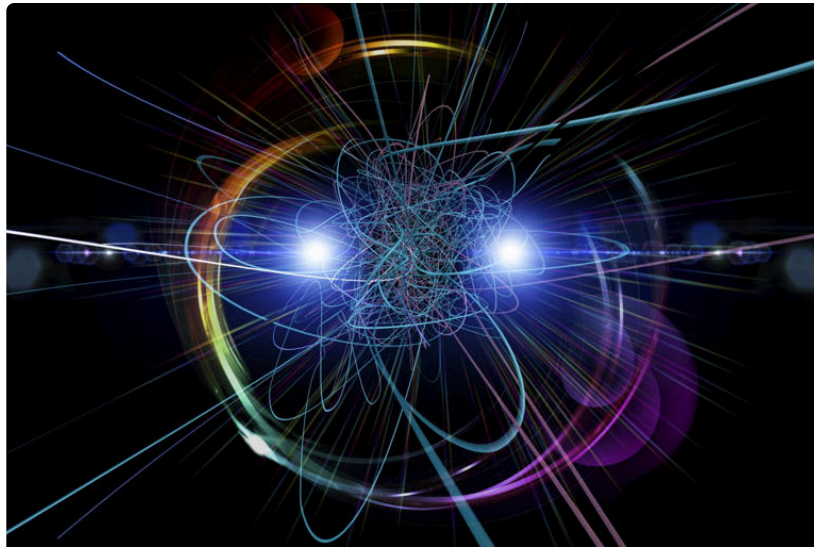
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Richard Feynman's 80-year-old quantum postulate has now been validated

Story by Karmela Padavic-Callaghan · 1w · 3 min read



 Feynman's "path integral" is a fundamental principle of quantum physics SCIENCE PHOTO LIBRARY/Getty Images

A foundational postulate of quantum physics has been directly verified by an experiment for the first time. Formulated by [Richard Feynman](#) nearly 80 years ago, it has long formed the backbone of much of modern physics.

Quantum particles are notoriously difficult to analyse as interacting with them [affects their properties](#). But researchers have developed powerful mathematical tools for predicting what a particle is likely to do under particular conditions.

One such tool is the "Feynman path integral", which posits that the most likely path a quantum particle will take between two points can be predicted by performing a calculation that sums all the possible paths between those points in a special way. Since Feynman proposed it in 1948, his path integral had made its way into textbooks and is now invaluable for many scientists, including physicists who study particles.

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